Assessment Brief

<table>
<tr><td colspan="2">Module Leader: Dr. Abayomi (Yomi) Otebolaku</td><td colspan="2">Level: 7</td></tr>
<tr><td colspan="2">Module Name: Programming Concepts and Practice</td><td colspan="2">Module Code:55-706555</td></tr>
<tr><td colspan="4">Assignment Title: Task 2: Stroke Data analytics</td></tr>
<tr><td>Individual</td><td>Weighting: 40%</td><td colspan="2">Magnitude: 1500</td></tr>
<tr><td>Submission date/time:
29-05-2025</td><td>Blackboard submission: Yes
Turnitin submission: Yes</td><td colspan="2">Format: Word, PowerPoint, source code, and digital media.</td></tr>
<tr><td>Planned feedback date:
19-06-2025</td><td>Mode of feedback: Verbal and written feedback</td><td colspan="2">In-module retrieval: No</td></tr>
<tr><td rowspan="2">In this assessment are students asked to consider:</td><td colspan="2">Inclusivity and accessibility</td><td>Yes</td></tr>
<tr><td colspan="2">Sustainability</td><td>Yes</td></tr>
<tr><td colspan="4"></td></tr>
<tr><td colspan="4"><u>Module Learning Outcomes</u>

● LO1: select appropriate programming techniques and data structures to develop effective software implementations of relatively complex systems using an appropriate programming language such as Python, Java, or C#.
● LO2: apply relevant program design strategies to the implementation of software applications using that programming language.
● LO3: design and implement well-engineered, domain specific software using that programming language.</td></tr>
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Assessment Brief

1. Introduction

According to the NHS, cardiovascular disease is a general term for conditions that affect the heart or blood vessels. It is usually caused by several factors such as build-up of fatty deposits in the arteries and increased risk of blood clots. It is also associated with damage to arteries and other vital organs such as brain, heart, kidneys, and eyes. Apart from leading to strokes, it can also lead to the death of the affected individuals. In the UK, for example, it is recognized as one of the main causes of death.

However, digital health is transforming not only how people's health is managed but also helping to prevent avoidable deaths such as those related to cardiovascular diseases. From translating health data and care into actions, it is gradually becoming an integral part of how healthcare providers (and patients themselves) manage health related problems. In addition, the availability of network enabled devices such as smartphone and wearable devices is enabling mobile medical apps and software that can provide support for clinicians to make clinical decisions using available data and artificial intelligence (AI) or machine learning (ML). Digital health can help to manage cardiovascular diseases and to minimize the number of deaths using people's vital signs or lifestyles to predict the likelihood of an impending fatal organ failures or death.

In this assignment, you have been provided with a dataset containing simulated 172,000 records of stroke related information about some anonymous patients. It is assumed you have been hired by a healthcare service provider as a data scientist. Part of your job is to analyze this dataset and then design and implement an application that can help clinicians monitor patients' vital signs and lifestyles such as blood glucose level, physical activities, stress, sleep hours, dietary habits, smoking habits, etc. to prevent an impending fatality.

The assignment will be assessed in two distinct phases, corresponding to coursework 1 and coursework 2, in line with the module's learning outcomes. The first assignment is linked with the second assignment in that the provided dataset will be used for both assignments. Both assignments are individual pieces of work, and your submission must be in the form of implemented modules and Jupyter notebook file, a 20-minute demo and a 5-page report.

2. Getting Started and General Specifications

The following tasks are to be performed in this assignment.

a. Implementing solution using modules and Object-Oriented Concepts

In this task, you will design and implement your solution using OOP concepts. You may choose to implement all or some key components of your solution using classes, methods, etc. As much as you can, throughout the entire implementation, please demonstrate your understanding of key object-oriented programming concepts that you have learnt.

	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
1	Ever Married	Work Type	Residence Type	Average Glucose Level	BMI	Smoking Status	Physical Acti	Dietary Habi	Alcohol Cons	Chronic Stress	Sleep Hours	Family Histo	Education Level	Income Level	Stroke Risk S	Region	Stroke Occurrence
2	1	Private	Rural	267.3	36.2	Formerly smoked	Sedentary	Non-Vegetar	0	0	11	0	Tertiary	Middle	88	South	0
3	0	Private	Rural	207.24	19.8	Never smoked	Light	Vegetarian	0	0	9	0	Secondary	High	56	East	0
4	0	Private	Urban	161.3	34.5	Never smoked	Sedentary	Mixed	1	0	3	0	Secondary	High	63	North	1
5	1	Never Worked	Urban	247.49	48.8	Never smoked	Active	Mixed	0	1	4	0	Tertiary	Middle	93	South	0
6	1	Government	Urban	116.57	31.3	Formerly smoked	Light	Mixed	0	1	9	0	Secondary	High	99	West	0
7	1	Government	Urban	220.62	49.9	Never smoked	Active	Non-Vegetar	0	1	7	0	No education	Low	56	North	0
8	0	Children	Urban	193.32	46.8	Never smoked	Sedentary	Vegetarian	0	0	5	0	Secondary	High	87	South	0
9	0	Never Worked	Urban	198.59	23.3	Never smoked	Light	Mixed	0	0	10	0	Primary	Low	5	North	0
10	0	Children	Rural	117.81	16.5	Unknown	Sedentary	Non-Vegetar	1	0	8	1	Secondary	Low	98	East	0
11	0	Government	Urban	289.09	37	Never smoked	Light	Vegetarian	0	0	8	0	Tertiary	Middle	94	East	0
12	1	Private	Urban	150.86	34.3	Never smoked	Moderate	Non-Vegetar	0	0	4	0	Primary	Middle	31	South	0
13	1	Private	Urban	254.9	49.5	Never smoked	Light	Non-Vegetar	0	0	6	1	Secondary	High	30	South	0
14	0	Self-employed	Rural	164.44	11.4	Never smoked	Sedentary	Non-Vegetar	0	0	10	0	Secondary	Low	72	South	0
15	1	Self-employed	Rural	142.49	15.2	Never smoked	Sedentary	Non-Vegetar	0	0	4	0	Primary	Middle	67	North	1
16	1	Government	Rural	218.1	36.5	Formerly smoked	Sedentary	Non-Vegetar	0	0	11	0	Secondary	Low	19	West	0
17	0	Private	Rural	245.06	10.9	Never smoked	Light	Vegetarian	0	0	4	0	Primary	Low	8	West	0
18	1	Private	Urban	266.21	18.9	Formerly smoked	Active	Vegetarian	0	0	6	0	Primary	Middle	53	East	0
19	1	Self-employed	Rural	221.45	34.5	Formerly smoked	Light	Non-Vegetar	0	0	10	0	Secondary	Low	63	North	0
20	1	Private	Rural	131.46	24	Never smoked	Sedentary	Vegetarian	0	1	4	0	No education	Low	100	West	0
21	1	Private	Urban	275.52	26.6	Never smoked	Active	Vegetarian	0	0	9	0	Primary	Low	10	East	0
22	1	Private	Rural	81.77	12.6	Never smoked	Sedentary	Vegetarian	0	0	11	0	Tertiary	Low	95	West	0
23	0	Private	Urban	150.87	24.1	Never smoked	Light	Non-Vegetar	0	0	3	0	Primary	Middle	31	South	0
24	0	Private	Urban	244.7	32.8	Formerly smoked	Sedentary	Non-Vegetar	0	0	6	0	Secondary	Low	65	South	0
25	0	Private	Urban	214.01	17.2	Formerly smoked	Sedentary	Mixed	0	1	12	0	Secondary	Low	16	West	0
26	0	Private	Urban	157.5	48.9	Never smoked	Sedentary	Vegetarian	0	0	8	0	Secondary	Low	77	North	0
27	0	Private	Urban	288.63	45.3	Never smoked	Light	Non-Vegetar	0	1	3	0	Primary	Low	77	South	0
28	0	Self-employed	Urban	177.89	32.5	Formerly smoked	Sedentary	Non-Vegetar	0	0	3	0	Tertiary	High	18	West	0
29	0	Government	Urban	169.66	10	Never smoked	Light	Vegetarian	1	0	11	0	Tertiary	Middle	88	South	0

Figure 1: This figure shows the features in the “data.csv”.

b. Exploratory Data Analysis (EDA)

You are required to write code to load the dataset and explore the data, checking missing data points, (and applying relevant cleaning techniques). All necessary data pre-processing must be executed via relevant python libraries. The EDA should be implemented as a module. The module should also include descriptive statistical analysis of the dataset (such as mean, median, standard deviation, variance, minimum, maximum, skewness, and kurtosis): choose a range of variables of your interest, find their frequencies, and dependencies through bar plots, grouped bar plots, pie-charts, box plots, scatter plots, etc. Interpret and report your conclusions.

In addition, explore the dataset to determine if the classes are balanced or not by producing the plot of the class distribution. If the classes are not balanced, use at least one relevant technique to address the class distribution issue. Finally, you will split the cleaned dataset into training and test datasets in preparation for training machine learning algorithms.

c. Statistical Feature Computation

Study the dataset and explore possible features that you need to extract for the next section(d) to train machine learning algorithms. It is your responsibility to identify relevant statistical features to be computed from the dataset if appropriate. If you conclude there is no need to extract features, please justify your decision. And if you extract features, you also need to justify why those features are important.

d. Predictions and Classifications

In this task, considering the features in the dataset, you will design, implement, train, and evaluate machine learning models for predicting the following classes:

1. Chronic Stress
2. Physical Activity
3. Income Level
4. Stroke Occurrence

For each of the above classes, you will implement and evaluate 3 classification models using machine learning algorithms. In your evaluation of these models, use metrics such as confusion matrix, precision, recall, accuracy, etc. present your results and discuss their implications. Visualization of your results such as precision, recall, accuracy, etc. by comparing the performance of implemented models is essential. For example, for each class in d (1-3), you will compare the performance of each of the three models and explain/justify why some algorithms perform better or worse than others.

e. Optional implementation

You may choose to implement instead of a text-based user interface in (c) above a Graphical User Interface (GUI) or any other extension (such as a database to persist the application's data). But discuss with your tutor to confirm what could be accepted as an extension. This will attract 5% of the total marks.

f. Report

You will write a report of your implementation, summarising your implementation decisions, justifications, and programming processes you have explored in the implementation of your deliverable. Provide instructions in your report on how to execute your application. Also, your report should contain structure of your program. It should also contain a reflection section explaining what you have learnt and how this has contributed to your professional development. Also reflect on what went well or not, and what you would have done better if given another opportunity to do this project again. Non submission of the report will attract a loss of 5% of the total marks. The report should be well structured with page numbers, etc., consisting of an abstract, introduction, design and implementation, discussion, reflection, conclusion, and references.

Suggested Structure of the Report:

- Cover page with your name, student number and title.
- Introduction which contains a short description of the context & method.
- Answers on the stated questions should be **well discussed.** For example, approach, and result/findings.
- All evaluation results should be presented in figures and tables (screenshots of Python result in Appendix)
- All plots, figures, and graphs must be numbered and clearly titled.
- Provide conclusions and recommendations.
- Please include a reference list, especially, to all libraries used in the implementation. You are required to use the APA style of referencing.

e. Demonstration

You are required to demonstrate (face-to-face) your deliverable showing and walking through how it meets the assessment criteria. Note that your live demonstration will not exceed 10 minutes. There will be question and answer session that will last for a maximum of 10 minutes.

3. Implementation

The implementation of the deliverable will be in modules. Three modules will be designed and implemented. The first module is the *load_dataset_module*, second module is the eda_module, and the third module is ui_module. The user interaction can be implemented using GUI (as an extension) or ordinary user-friendly text-based UI. These three components of the system should be seamlessly integrated so that during your walkthrough it should be clear how they all interact. Your implementation should be in Python. The modules will be implemented as python scripts (.py files) and your main function should be implemented as a Jupyter notebook file (. ipynb). If you are implementing a GUI as an extension, please feel free to use any Python GUI library.

4. Pay attention to the following requirements

a) In-module retrieval is **NOT** available for this assessment.

b) This assignment is an individual piece of work, and your submission must be in the form of **three module files (.py) and a Jupyter Notebook file (.ipynb)**. The Jupyter notebook contains the main function integrating the 3 modules. **If the submission is not in the specified format, 5% of the total marks will be deducted as penalty.**

c) You will submit a report. The report should provide analysis and design artefacts of the problem being solved, justification for your design decisions and pseudocodes of the functions you have designed and developed. It should explain the relationships between the modules. A good report should provide evidence of critical analysis of the implemented system. Even if your application does not work correctly, you should still submit the report explaining what you have done, what works and what has not worked. Please note that you are required to submit your report both to the Turnitin submission point as well as the main assessment submission point. Failure to submit your report to both will attract a 5% penalty.

d) Any evidence of collusion/plagiarism will be penalised! If there is some doubt about the authenticity of a particular piece of work, then the person submitting it will be expected to defend such work, including reasons for the programming decisions taken. You must document with references any use of libraries or existing code in your report.

e) Appropriate use of variable names for clearer understating is desirable. Adequate commenting of your code for easier understanding is also desirable.

f) You are not allowed to use AI tools to generate code and submit such as your own implementation. This will be regarded as an academic misconduct and appropriate penalty will be applied as deem fit by the academic misconduct panel.

5. Submission Process

Your assignment should be submitted electronically through the module's Blackboard site as a single ZIP file that contains all source codes and report, not more than 5 pages. Check your upload to ensure you have submitted the correct files successfully as any issues will not be considered after the deadline. Please, note that the submission deadline is **Thursday, 29th, May 2025, 3:00 pm.**

- **Check your upload to ensure you have submitted the correct files successfully as we will not consider any excuse after the deadline.**
    - **Note that any wrong submission or submission of wrong file or deliverable will be graded as 0. There is no room for sending deliverable file via email.**
    - **Please ensure that your correct files are submitted as we will not entertain any request for mistaken submission after the deadline.**
- **Note that late submission will attract penalty. The penalty is capping your mark to 50% or 0 depending on how late the submission is.**
- **You must also check your report's similarity score using Turnitin on the Blackboard before final submission. Please do not submit any report with a similarity score higher than 20%. Otherwise, the University may penalize you for plagiarism or collusion.**

6. Assessment Criteria

This assignment will be assessed through the report, testing of implementation and video demonstration of the submitted deliverable. The video demo should show how your solution meets the assessment criteria. In general, the coursework will be assessed against the Learning Outcomes (LOs) using a set of assessment criteria. This set of assessment criteria allows assessing how successful you have met the LOs. In order to ensure consistent use of the relevant criteria, the assessment criteria are summarised in the following assessment matrix and grid. This is an indicator of how the marks will scale across each category of the learning outcomes it covers.

Note that the University's categorical grading descriptor (in Figure 1 below) will be used to determine your grade in this assessment. The marking scheme embeds the concept of extended work by rewarding only the highest marks to those who demonstrate evidence of independent investigation, learning, and thought. Thus, to achieve top grades, you will need to go beyond the materials presented in lectures and labs and undertake some of your own research (i.e., read and discuss related materials).

Table 1: Assessment Matrix

Assessment 2 Criteria	Weights	Learning Outcome Covered
Clear understanding of relevant programming concepts such as Object-Oriented Programming concepts of classes, encapsulation, objects, methods, inheritance, custom module, functions, parameters, and arguments, etc.	10%	LO1, LO2, LO3
Use of python (data science) libraries for exploratory data analysis (EDA) and statistical computation.	10%	LO3

Use of python machine learning library training and fitting predictive models.	10%	LO3
Subject knowledge/Video Demonstration/Report/Deliverable.	5%	LO2, LO3
Optional Extension	5%	LO1, LO2, LO3

Note that extension is 5%.

Table 2: Assessment Marking Grid

Fail (<50%)	**Pass (50-59)**	**Merit (60-69)**	**Distinction (70% +)**
Understanding of relevant programming concepts (/10)			
No evidence of understanding and use of OOP (classes, objects, methods, inheritance, etc.), function definition, parameter and argument passing. **Nothing is submitted.**	**Evidence of clear and consistent understanding of the OOP (classes, objects, methods, inheritance, etc.), function definition, parameters, and argument passing.** **Evidence of practical solution.**	**Very good and appropriate OOP (classes, objects, methods, inheritance, etc.), definition of functions, parameters and argument passing.**	**Exceptional understanding and creative use of programming solutions. OOP (classes, objects, methods, inheritance, etc.), function definition, parameter passing etc. Robust implementation of deliverable.**
Use of python libraries for exploratory data analysis (EDA) and statistical feature computation (/10)			
No evidence of the use of Python libraries such as NumPy, pandas, matplotlib, seaborn etc. Not able to apply appropriate python libraries. **No submission.**	**Clear and good evidence of the use and application of python libraries (such as NumPy, pandas, matplotlib and seaborn) with some correct and expected outputs. But some minor issues with outputs.**	**Very good understanding and good implementation using python libraries to implement some of the functionality of the system with justifications and correct outputs. Program executes and produces expected**	**Excellent understanding and implementation of relevant python libraries, such as NumPy, pandas, matplotlib and scikit-learn with outstanding results. excellent user interaction using menu-driven interface or a GUI, etc.**
Use of python (data science) libraries for building predictive models (/10)			
No evidence of the use of Python libraries such as NumPy, pandas, especially scikit-learn. Not able to apply appropriate python libraries. Limited or no interpretation of evaluation results.	**Clear and good evidence of the use and application of python libraries (such as NumPy, pandas, matplotlib and scikit-learn) with some correct and expected outputs. But some minor issues with outputs. Good**	**Very good understanding and good implementation using python libraries to implement some of the functionality of the system with justifications and correct outputs. Program executes and produces expected. Very good**	**Excellent understanding and implementation of relevant python libraries, such as NumPy, pandas, matplotlib and scikit-learn with outstanding results. excellent user interaction through a GUI, etc. Excellent interpretations of evaluation results.**

No submission	**interpretation of evaluation results.**	**interpretation of evaluation results.**	
Knowledge of subject// Demonstration/Report/Deliverable (/5)			
Report lacking good structure, no personal reflection, no description of the deliverable or explanation and justification of decisions. Poor use of language. No/Poor video demonstration.	**Good structure, evidence of personal reflection on what went well or not. Good recommendations were made. Good justification for design and implementation decisions. Good use of language. Good video demonstration.**	**Very good structure, evidence of personal reflection on what went well or not. Very good recommendations were made. Very good justification for design and implementation decisions. Very good video demonstration. Good use of language.**	**Excellent structure, excellent personal reflection on what went well or not. Excellent recommendations were made. Good justification for design and implementation decisions. Excellent video demonstration. Excellent use of language. Evidence of innovation in the deliverable, e.g., excellent user interaction using user friendly menu-driven user interface a GUI, etc.**
Extension (/5)			
No extension is submitted. Extension submitted but does not work.	Extension submitted, and demonstrated, works as intended but has some limitation, e.g. A GUI that does not aesthetic or not friendly.	Extension submitted, demonstrated, works as intended, no noticeable limitation.	Extension submitted, demonstrated works as intended, no limitations, professional level extensions.

University Grade Descriptor (UGD)

Level 7: Generic grade descriptor: relationship of grades of achievement to percentage mark ranges and categorical grades (CG)

Class	Category	Mark range	%	General Characteristics
Distinction	Exceptional Distinction	93 - 100	96	Exceptional breadth and depth of knowledge and understanding evidenced by own independent insight and critical awareness of relevant literature and concepts at the forefront of the discipline; evidence of extensive and appropriate independent inquiry operating with advanced concepts, methods and techniques to solve problems in unfamiliar contexts; Cogent arguments and explanations are consistently provided using a range of media demonstrating an ability to communicate effectively in a variety of formats using a sophisticated level of the English language in an eloquent and professional manner to both technical and non-technical audiences; a sustained academic approach to all aspects of the tasks is evidenced; academic work extends boundaries of the disciplines and is beyond expectation of the level and may achieve **publishable or commercial standard**.
Distinction	High Distinction	85 - 92	89	Excellent knowledge and understanding evidenced by some **clear independent insight and critical awareness of relevant concepts some of which are at the forefront of the discipline**; evidence of appropriate independent inquiry operating with core concepts, methods and techniques to solve complex problems in mostly familiar contexts; Arguments and explanations are provided that is well-supported by the literature and in some cases uses a range of media demonstrating an ability to communicate effectively in a limited number of formats using own style that is suited to both technical and non-technical audiences; a sustained academic approach to most aspects of the tasks is evidenced; **one or more aspects of the academic work is beyond the prescribed range** and evidences a competent understanding of all of the relevant taught content.
	Mid Distinction	78 - 84	81	
	Low Distinction	70 - 77	74	
Merit	High Merit	67 - 69	68	Very good knowledge and understanding is evidenced as the student is **typically able to independently relate taught facts/concepts together some of which are at the forefront of the discipline**; evidence of some competent independent inquiry operating with core concepts, methods and techniques to solve familiar problems; Arguments and explanations are provided that are typically supported by the literature and in some cases may challenge some received wisdoms; competently uses all taught media and communication methods to communicate effectively in a familiar settings; an academically rigorous approach applied to some aspects of the tasks is evidenced; some beyond the prescribed range, may rely on set sources to advance work/direct arguments; demonstrates autonomy in approach to learning.
	Mid Merit	64 - 66	65	
	Low Merit	60 - 63	62	
Pass	High Pass	57 - 59	58	Satisfactory knowledge and understanding of the area of study **balanced towards the descriptive rather than critical or analytical and mostly confined to concepts that are not at the forefront of the discipline**; evidence of some independent reading and research to advance work and inform arguments and approaches; Arguments and explanations are limited in range and depth although some are adequately supported by the literature albeit descriptively rather than critically; competently uses at least one taught media and communication method to communicate appropriately in familiar settings; although the approach applied to some aspects of the tasks may lack academic rigour, there are some clear areas of competence within the prescribed range. Relies on set sources to advance work/direct arguments and communicated in a way which shows clarity but structure may not always be coherent.
	Mid Pass	54 - 56	55	
	Low Pass	50 - 53	50	
Fail	Borderline Fail	40 - 49	45	Knowledge and understanding is insufficient as the **student only evidences an understanding of small subset of the taught concepts and techniques; fails to make sufficient links between known concepts and facts** to adequately solve relevant aspects of the brief/problem; little ability to independently select and evaluate reading/research with almost total reliance on set sources and unsubstantiated arguments/methods; communication/presentation may be competent in places but fails to demonstrate structure, clarity and/or focus; inability to adequately define problems and make reasoned judgements; the general approach to tasks lacks rigor and competence.
	Mid Fail	30 - 39	35	
	Low Fail	20 - 29	25	
Fail	Very Low Fail	6-19	10	Knowledge and understanding is highly insufficient as the student is **unable to evidence any meaningful understanding of taught concepts or methods**; very limited evidence of reading and research to advance work; inadequate technical and practical skills as the student is unable to use and apply such skills to address problems or make judgements; limited or lack of understanding of the boundaries of the discipline and does not question received wisdom; approach to learning lacks autonomy and approach to tasks is not sustained; inability to communicate coherently.
Zero	Zero	0-5	0	Work not submitted, work of no merit, penalty in some misconduct cases.

Figure 4: The University Grade Descriptor for Level 7. Your final class, category and grade for the module will be determined using the UGD.

Artificial Intelligence and Academic Integrity – AI&AI

It is important you do not use AI tools to generate an assignment and submit it as if it were your own work. Our regulations state:

Contract cheating/concerns over authorship: This form of misconduct involves another person (or artificial intelligence) creating the assignment which you then submit as your own. Examples of this sort of misconduct include buying an assignment from an ‘essay mill’/professional writer; submitting an assignment which you have downloaded from a file-sharing site; acquiring an essay from another student or family member and submitting it as your own; attempting to pass off work created by artificial intelligence as your own. These activities show a clear intention to deceive the marker and are treated as misconduct.

Further guidance is available here: https://blogs.shu.ac.uk/assessment4students/preparing-to-submit-work/#AI

University Study Rules and Regulations

See Grades and Marks > University grade descriptors

https://www.shu.ac.uk/myhallam/university-life/university-rules-and-regulations/study

Assessment Regulations – staff page

University Grade Descriptors: This version also includes guidance on UGD3 and FAQs

[Assessment, Progression and Awards webpage]

Please also refer to: BTE Assessment page for other college guidance

https://sheffieldhallam.sharepoint.com/sites/3020/SitePages/Assessment-information.aspx